

ESP32-CAM Monitoring System: Project Roadmap

Phase 1: Planning & Architecture [COMPLETE]

- ☒ ~~Define system behavior: Wake up every 15 mins, record 10s CIF@30fps, sleep (16 mins active runtime per day).~~
- ☒ ~~Design local storage: 3-day rolling buffer on the SD card, automatically clearing the oldest day.~~
- ☒ ~~Design communication: Utilize HTTP POST for delivery only when a file needs to be permanently saved.~~
- ☒ ~~Design UI/UX: Web Dashboard for a 24-hour timeline and Cloud Backup (10 permanent "Safety Deposit Box" slots per user).~~
- ☒ ~~Define Failsafe: Ensure manual SD card extraction works if the network fails.~~

Phase 2: Component Testing [COMPLETE]

- ☒ ~~Establish secure global tunnel (Tailscale) to bypass network firewalls.~~
- ☒ ~~Configure Linux server backend (Django views.py & urls.py routing).~~
- ☒ ~~Verify HTTP POST reception using simulated data.~~
- ☒ ~~Test ESP32 dual-core CIF 30fps recording.~~
- ☒ ~~Test ESP32 Wi-Fi initialization and SD card isolation.~~

Phase 3: Core Development & Integration [IN PROGRESS]

- ☒ ~~**Hardware (ESP32):** Implement deep sleep loop (wake every 15 mins).~~
- ☒ ~~**Hardware (ESP32):** Write the rolling 3-day buffer logic for local SD file management.~~
- ☒ ~~**Backend (Django):** Handle 10 cloud backup slots and API routing for permanent saves.~~
- ☐ **Frontend (Web):** Enmanuel to develop the user dashboard for the 24-hour timeline.
- ☐ **Integration:** End-to-end testing of the fully automated system.

Phase 4: Hardware Enclosure [PENDING]

- ☐ Source or design 3D printable chassis.
- ☐ Print chassis and assemble final hardware components.
- ☐ Final deployment and field testing.